HARD · ADVANCED MATH

# Nonlinear functions

# 05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

**Q1** ADVANCED MATH

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

A. $\frac{a}{4}$

B. $a$

C. $4a$

D. $6a$

Q2

GRID-IN ADVANCED MATH

Function $f$ is a quadratic function where $f{-20} = 0$ and $f{-4} = 0$. The graph of $y = fx$ in the $xy$-plane has a vertex at $r, -64$. What is the value of $r$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN ADVANCED MATH

$$fx = 4x^2 - 50x + 126$$

The given equation defines the function $f$. For what value of $x$ does $fx$ reach its minimum?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN ADVANCED MATH

The function g is defined by $gx = xx - 2x + 6^2$. The value of $g\left(7 - w\right)$ is 0, where $w$ is a constant. What is the sum of all possible values of $w$?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 ADVANCED MATH

The function f is defined by $f(x)=(x+3)(x+1)$ . The graph of f in the xy-plane is a parabola. Which of the following intervals contains the x-coordinate of the vertex of the graph of f ?

A. $-4<x<-3$

B. $-3<x<1$

C. $1<x<3$

D. $3<x<4$

Q6 ADVANCED MATH

$$y = 576^{2x + 2}$$

The graph of the given equation in the $xy$-plane has a $y$-intercept of $r, s$. Which of the following equivalent equations displays the value of $s$ as a constant, a coefficient, or the base?

A. $y = 331{,}776^{x + 1}$

B. $y = 24^{4x + 4}$

C. $y = \frac{1}{24} 24^{4x + 5}$

D. $y = \frac{1}{576} 576^{2x + 3}$

Q7 ADVANCED MATH

The function $f$ is defined by $f x = a 2.2^x + 2.2^b$, where $a$ and $b$ are integer constants and $0 < a < b$. The functions $g$ and $h$ are equivalent to function $f$, where $k$ and $m$ are constants. Which of the following equations displays the $y$-coordinate of the $y$-intercept of the graph of $y = f x$ in the $xy$-plane as a constant or coefficient?

I. $g x = a 2.2^x + k$
II. $h x = a 2.2^x + m$

A. I only

B. II only

C. I and II

D. Neither I nor II

Q8

ADVANCED MATH

A machine launches a softball from ground level. The softball reaches a maximum height of $51.84$ meters above the ground at $1.8$ seconds and hits the ground at $3.6$ seconds. Which equation represents the height above ground $h$, in meters, of the softball $t$ seconds after it is launched?

A. $h = -t^2 + 3.6$

B. $h = -t^2 + 51.84$

C. $h = -16t - 1.8^2 - 3.6$

D. $h = -16t - 1.8^2 + 51.84$

**Q9** GRID-IN ADVANCED MATH

A rectangle has an area of 155 square inches. The length of the rectangle is 4 inches less than 7 times the width of the rectangle. What is the width of the rectangle, in inches?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, $d$ and $g$ are unique positive constants. When the equation is graphed in the $xy$-plane, how many distinct $x$-intercepts does the graph have?

A. 4

B. 3

C. 2

D. 1